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Maths puzzles for early finishers

# Early Finisher Puzzles

## Primary 3

### Semester 1

Terms 1 and 2

20 single-page puzzles
Almost every puzzle checks itself
One star and two star pages
Teacher answer key included

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# Teacher notes (DRAFT)

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## What this is

Twenty single-page maths puzzles for Primary 3 pupils who finish their work early. This is Semester 1 of two booklets (Terms 1 and 2). Each page stands alone, so you can hand out one page or keep a tray of pages. The pupil needs no help to start.

## Which terms

The pages are grouped by the order in which Primary 3 topics are typically taught: this booklet follows Terms 1 and 2, the other booklet follows Semester 2 (Terms 3 and 4). Schools vary, so check your scheme of work: use a page whenever its topic has been taught, from either booklet.

## Stars

One star pages practise a topic as it is usually taught (10 pages). Two star pages use the same topics with two steps or two clues to combine (10 pages).

## How the pages check themselves

The boxed strip at the foot of each page tells the pupil how to check. The check shows that something is wrong. It does not give the answer, and it asks the pupil to see whether the answer makes sense. Every answer box and every letter box must be filled: a box left blank gives no letter, and the check cannot see it.

- **Code word.** The answers give letters that spell a word. The strip gives no hint of the word. Each word can be spelt from its letters in one way only, so answers in the wrong boxes never make another word, and one or two wrong answers very rarely do. A box left empty gives no letter and is not seen by the check.
- **Colour by code.** Each answer is found in a key of five numbers and the part is coloured with that crayon. The pupil counts the parts of each colour and compares with the box on the page; matching parts must share a colour.
- **Route.** The line must reach FINISH through the stated number of boxes, and every box on it must follow the rule on the page.
- **Letters in an order, a grid or a choice.** Each number, card, picture or choice carries a letter; the letters, read in the order asked, must spell a word. Where one item is left out, its letter is a decoy that is not part of the word.
- **Loop.** The numbers on touching tiles must match all the way round the loop.

Marking is not needed. To spot-check, look only at the check strip.

## How the checks were tested

A script tried every single wrong answer and every wrong arrangement on every page. Each check rejected at least 95% of them. A wrong answer that spells any of about 262 000 English words counts as not rejected. Two wrong answers at once were not all tested.

# Teacher notes: using the pages (DRAFT)

## Using the pages

- **Time.** Most pages take 5 to 15 minutes; the colour-by-code pages (S1-14) take longer because pupils colour a picture. Pupils need their own crayons (five colours: yellow, red, blue, green, orange). For a pencil-only copy the pupil writes Y, R, B, G or O in each part; the tally and the matching parts are checked the same way. Light crayons work best: a dark crayon hides the number in the part on a photocopy.
- **If the check fails.** The pupil goes back, looks for the slip, and asks a friend or you if still stuck.
- **Stars.** One star: one skill, as usually taught. Two stars: the same topics with two steps or two clues to combine.
- **Drawings.** Pictures of rulers, scales, clocks, coins and notes are original drawings, not to scale unless a page says so. Coins and notes are inspired by Singapore money, not actual size (smaller than real) and in black and white.

## Photocopying

All pages are black and white with solid lines, so they copy cleanly. Use normal toner, not a light setting: on the page with coins and notes (Money Sums, S1-20) a light setting can fill in the small word cents on the coins. Each page is one side only. Pupils write on the page: the boxes are large for young hands. Use A4 and print at 100% (actual size).

Pictures are not actual size. Real $2 notes are purple, $5 green, $10 red.

## What a printed page cannot check

Real measuring (a ruler, a jug, a scale, a stopwatch) cannot be done on paper: the pages show drawn scales, containers, dials and stopwatch faces to read, and the pupil reads them. The idea of an angle is met by comparing drawn angles with a right angle. Drawing perpendicular and parallel lines (Draw the Lines, S2-16) is checked only through the letter where the drawn line ends, so look at the lines if you wish. Mental calculation cannot be enforced on a page: the route and code pages ask for answers, not for the method. Figures with marked lengths are drawn to scale on the page, but pictures are not actual size.

# Contents: Semester 1


| Page | Code | Puzzle | Tier |
|---|---|---|---|
| S1-01 | P3-01 | Place Value | 1 star |
| S1-02 | P3-02 | Words and Numbers | 1 star |
| S1-03 | P3-03 | Digit Cards | 2 stars |
| S1-04 | P3-04 | Compare and Order | 2 stars |
| S1-05 | P3-05 | Number Patterns | 2 stars |
| S1-06 | P3-06 | Big Sums | 1 star |
| S1-07 | P3-07 | Mental Pairs | 1 star |
| S1-08 | P3-08 | Missing Digits | 2 stars |
| S1-09 | P3-09 | Tables Maze | 1 star |
| S1-10 | P3-10 | True or False Tables | 1 star |
| S1-11 | P3-11 | Remainders | 1 star |
| S1-12 | P3-12 | Fact Families | 2 stars |
| S1-13 | P3-13 | Multiply and Divide | 2 stars |
| S1-14 | P3-14 | Colour the Fish | 2 stars |
| S1-15 | P3-15 | Equal Fractions | 1 star |
| S1-16 | P3-16 | Simplest Form | 1 star |
| S1-17 | P3-17 | Fraction Gaps | 2 stars |
| S1-18 | P3-18 | Fraction Order | 2 stars |
| S1-19 | P3-19 | Fraction Sums | 2 stars |
| S1-20 | P3-20 | Money Sums | 1 star |


Pages are independent. The footer code shows the page in this booklet and, in brackets, its number in the full set of 40 (P3-01 to P3-40).

## What the booklet covers

**Whole Numbers 1.1** counting in hundreds/thousands **S1-01, S1-05**

**Whole Numbers 1.2** number notation, representations and place values (thousands, hundreds, tens, ones) **S1-01, S1-03**

**Whole Numbers 1.3** reading and writing numbers in numerals and in words **S1-02**

**Whole Numbers 1.4** comparing and ordering numbers **S1-03, S1-04**

**Whole Numbers 1.5** patterns in number sequences **S1-05**

**Addition and Subtraction 2.1** addition and subtraction algorithms (up to 4 digits) **S1-06, S1-08**

**Addition and Subtraction 2.2** mental calculation involving addition and subtraction of two 2-digit numbers **S1-07**

**Multiplication and Division 3.1** multiplication tables of 6, 7, 8 and 9 **S1-09, S1-10, S1-12, S1-14**

**Multiplication and Division 3.2** multiplying and dividing within the multiplication tables **S1-09, S1-10, S1-12, S1-14**

**Multiplication and Division 3.3** division with remainder **S1-11**

**Multiplication and Division 3.4** multiplication and division algorithms (up to 3 digits by 1 digit) **S1-13**

**Multiplication and Division 3.5** mental calculation involving multiplication and division within the multiplication tables **S1-09, S1-10, S1-14**

**Fractions 1.1** equivalent fractions **S1-15, S1-17, S1-18**

**Fractions 1.2** expressing a fraction in its simplest form **S1-16**

**Fractions 1.3** comparing and ordering unlike fractions with denominators of given fractions not exceeding 12 **S1-18**

**Fractions 1.4** writing the equivalent fraction of a fraction given the denominator or the numerator **S1-17**

**Fractions 2.1** adding and subtracting two related fractions within one whole with denominators of given fractions not exceeding 12 **S1-19**

**Money 1.1** adding and subtracting money in decimal notation **S1-20**

Outcomes practised in the other booklet: Length, Mass and Volume 1.1; Length, Mass and Volume 1.2; Length, Mass and Volume 1.3; Time 2.1; Time 2.2; Time 2.3; Area and Perimeter 1.1; Area and Perimeter 1.2; Area and Perimeter 1.3; Area and Perimeter 1.4; Angles 1.1; Angles 1.2; Perpendicular and Parallel Lines 2.1; Perpendicular and Parallel Lines 2.2; Bar Graphs 1.1; Bar Graphs 1.2.